

Fruit fly assay report

Session: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_131245

Run dir: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_131245

Raw video:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_131245/raw_video.mp4

Background:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_131245/background_transformed_snapshot.png

Calibration:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_131245/calibration_snapshot.json

Duration (s): 9.8

Frames processed: 50

Tracking mode: stitch

Mask blobs as flies: True

Assay tubes: 4

Track fragments detected: 29

Unique threshold crossings: 4

0 s to 9.8 s comparison window

0 s to 9.8 s x-displacement summary
Columns 1-6 of 9 | Rows 1-4 of 4

assay_tube_index	flies_compared	mean_abs_dx_px	median_abs_dx_px	mean_signed_dx_px	comparison_target_s
1	1	147.593	147.593	147.593	9.8
2	4	63.031	45.779	-30.184	9.8
3	1	0.18	0.18	0.18	9.8
4	1	115.9	115.9	115.9	9.8

0 s to 9.8 s x-displacement summary
Columns 7-9 of 9 | Rows 1-4 of 4

mean_abs_dx_mm	median_abs_dx_mm	mean_signed_dx_mm
14.26	14.26	14.26
6.67	4.844	-3.194
0.019	0.019	0.019
12.667	12.667	12.667

Threshold crossing summary
Columns 1-5 of 5 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	number_of_unique_threshold_crossings	fraction_crossing_by_10s	first_crossing_time_s
1	10	1	0.1	7.6
2	7	0	0.0	nan
3	4	0	0.0	nan
4	8	3	0.375	3.8

Threshold crossing events
Columns 1-6 of 8 | Rows 1-4 of 4

unique_event_id	assay_tube_index	display_id	crossing_time_s	crossing_x_px	crossing_y_px
1	1	8	7.6	318.367	261.68
2	4	3	3.8	964.865	240.427
3	4	2	4.6	896.549	241.526
4	4	6	5.0	970.311	238.509

Threshold crossing events
Columns 7-8 of 8 | Rows 1-4 of 4

threshold_distance_px	threshold_distance_mm
501.0	66.912
500.0	68.143
500.0	68.143
500.0	68.143

Velocity summary
Columns 1-6 of 6 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	mean_speed_mm_s	max_speed_mm_s	mean_vertical_velocity_mm_s	max_vertical_velocity_mm_s
1	10	6.107	17.078	4.898	17.047
2	7	2.832	8.822	2.203	8.759
3	4	2.253	15.575	-0.561	8.674
4	8	8.762	20.37	7.071	20.333